

M.U.L.A.N.: Thermodynamically Consistent Propellantless Propulsion via Asymmetric Nanoconfinement, Covalent Strain, and Optical Upconversion

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Abstract

We present a fully thermodynamically consistent propellantless thrust concept based on the asymmetric nanoconfinement and mechanical pre-straining of polar molecules (H_2O) within a refractory plasmonic lattice. The MULAN (Momentum Uplifting via Light Amplification in Nanoconfinement) v3 engine bypasses fast vibrational relaxation (1–2 ps) by utilizing a mechanical-optical pumping scheme. The architecture introduces a three-stage cascade: near-infrared (NIR) input photons are frequency-shifted via upconversion microspheres, routed through a unidirectional nano-hollander (photonic diode), and injected into geometrically constrained, pre-strained O-H covalent bonds. Thermal fluctuations from a hot refractory reservoir (Ta_4HfC_5 , $T_{melt} \approx 4200$ K) continuously recharge the bond strain. Spatial asymmetry is enforced via a passive photonic nano-collet, forcing highly directional photon emission. Momentum conservation is preserved, but a dimensionless amplification coefficient $K = p_{out}/p_{in} > 1$ is enabled via thermal assistance without violating the second law of thermodynamics. At a spacecraft power level of 30 W and active areas above 0.13 m^2 , MULAN v3 provides a scalable, multi-decade propellantless propulsion alternative to conventional ion thrusters.

1 Introduction

Classical photon thrusters are governed by the fundamental momentum of light:

$$F = \frac{P_{in}}{c} \quad (1)$$

where P_{in} is the net emitted optical power. In the original MULAN concept, localized plasmonic gain was coupled directly to thrust, causing a scaling paradox. MULAN v3 resolves this by treating the active molecular medium not as an energy multiplier, but as a *thermo-mechanical quantum spatial gate*.

The engine extracts thermal energy from a hot nanostructured reservoir and converts it into directed momentum through a pre-strained molecular lattice. We introduce the dimensionless **momentum amplification factor**:

$$K = \frac{p_{out}}{p_{in}} = \frac{\lambda_{in}}{\lambda_{out}} = 1 + \frac{P_{thermal_extract}}{P_{laser}} \quad , \quad (2)$$

where the emitted photon is blue-shifted ($\lambda_{out} < \lambda_{in}$) by absorbing thermal energy stored in the strained intramolecular bonds. The second law of thermodynamics sets the upper limit $K_{max} = T_{hot}/T_{cold}$, derived from Carnot efficiency considerations for a photon heat engine operating via sub-wavelength confinement.

2 Physical Model and Engineering Solutions

2.1 Geometric Nanoconfinement and Covalent Strain Pumping

In unconstrained liquid or gas phases, the O-H bond exhibits ultra-fast vibrational relaxation (1–2 ps), rendering direct infrared driving inefficient due to instant thermalization into lattice phonons. MULAN v3 bypasses this by lock-key spatial confinement. The geometry of the Ta_4HfC_5 nanolattice is engineered to rigidly clamp the oxygen atom of the H_2O molecule, forcing the hydrogen atoms into a structurally deformed, pre-strained state where the equilibrium distance r_e is artificially extended ($r_{stressed} > r_e$).

The O-H bond potential under strain is modeled via the modified Morse potential:

$$U(r) = D_e \left(1 - e^{-a(r-r_e)}\right)^2 - U_{stress}(\Delta r) \quad (3)$$

where $D_e = 7.688 \times 10^{-19}$ J, $r_e = 96$ pm, $a = 2.2 \times 10^{10}$ m⁻¹, and the baseline spring constant $k \approx 744$ N/m. The mechanical strain energy U_{stress} is continuously replenished by the high-temperature reservoir via phonon-molecular coupling. High-energy thermal fluctuations from the heavy tantalum/hafnium lattice continuously pump the pre-strained bond, maintaining it in a high-energy metastable state.

2.2 The Three-Stage Cascade: Upconversion, Hollander Gate, and Triggering

To efficiently trigger radiative relaxation without driving deep electronic transitions, a three-stage optical cascade is integrated directly into the input plane:

1. **Upconversion Microspheres:** Incoming NIR laser light ($\lambda = 1064$ nm) passes through a layer of upconversion dielectric microspheres. Through multi-photon absorption, these structures upconvert the low-energy IR photons into higher-frequency visible/UV trigger photons.

2.3 Passive Photonic Nano-Collet and Directional Emission

To prevent isotropic photon emission, the nanolattice terminates in a passive photonic nano-collet—a dielectric/plasmonic metasurface operating as a high-directivity sub-wavelength Yagi-Uda antenna array. This constrains the emitted solid angle $\Delta\Omega$, ensuring maximum asymmetric momentum transfer to the spacecraft hull with a directional efficiency η .

2.4 Refractory Plasmonic Substrates: TiN and Ta_4HfC_5

To withstand severe ohmic losses and maximize the Carnot bound, the lattice utilizes ultra-high-temperature ceramics (UHTCs). While Titanium Nitride (TiN, $T_{melt} \approx 2900^\circ\text{C}$) serves as a baseline, we employ **tantalum hafnium carbide** Ta_4HfC_5 ($T_{melt} \approx 4200$ K). Operating at $T_{hot} \approx 4500^\circ\text{F}$ (2750 K) via active/pulsed states yields a theoretical Carnot bound of $K_{max} \approx 15$ against a deep space background ($T_{cold} = 300$ K).

2.5 Thermodynamic Consistency and Net Thrust

The net thrust density per unit area is given by:

$$F_{area} = \eta \cdot K \cdot \frac{P_{laser}}{c} \quad (4)$$

where P_{laser} is the incident laser power density. Because the additional energy is drawn from the thermal capital of the UHTC reservoir ($P_{out} = P_{laser} + P_{thermal}$), total momentum is strictly conserved across the closed system, resolving the scaling paradox of prior versions.

3 Numerical Results: Scaling with Area and Power

We analyze a CubeSat configuration: 30 W total electrical power, converted to laser light with 50% efficiency (15 W net optical power P_{total}). The laser power density scales inversely with the chip area: $P_{laser} = P_{total}/A_{chip}$.

Table 1: Theoretical thrust for Ta_4HfC_5 system at 30 W electrical input ($\eta = 50\%$, $K = 15$)

Chip area (cm ²)	Chip size	Power density (W/mm ²)	Total Thrust (N)	Comparison to ion thruster
1	1 × 1 cm	0.15	0.75	1300× weaker
100	10 × 10 cm	0.0015	75	13× weaker
400	20 × 20 cm	0.000375	300	3× weaker
900	30 × 30 cm	0.000166	675	1.5× weaker
1296	36 × 36 cm	0.000115	972	1.03× weaker
1600	40 × 40 cm	0.000094	1200	1.2× stronger

For an integrated 1 × 1 m² panel (10,000 cm²), the total thrust reaches ≈ 7500 N = 7.5 mN, outperforming micro-ion systems at identical power inputs without mass depletion.

3.1 Break-even area

Setting $F_{MULAN} = F_{ion}$ (where $F_{ion} \approx 1000$ N at 30 W) yields the equilibrium area A_{eq} :

$$A_{eq} = \frac{F_{ion} \cdot c}{\eta \cdot K \cdot P_{laser}} \cdot A_0 \approx 0.13 \text{ m}^2 \quad (36 \times 36 \text{ cm}) \quad (5)$$

Above this threshold, the spatial thermal extraction area compensates for low baseline photon momentum, allowing propellantless scaling.

4 Thermal Management: Pulsed and Hybrid Operation

Dissipating accumulated waste heat at high power densities requires managed thermal duty cycles.

4.1 Pulsed Mode

Operating the laser trigger at a low duty cycle $D = t_{on}/t_{period}$ (e.g., $t_{on} = 0.1$ s, $t_{period} = 100$ s, $D = 0.001$) maintains peak localized thrust while reducing the average thermal load proportionally, preventing lattice degradation.

4.2 Hybrid Mode (Radiator + Pulses)

Integrating a passive radiator area $A_{rad} \approx 4 \times A_{chip}$ allows steady-state operation. At $T_{rad} = 400$ K, the radiated cooling capacity is $\sigma T^4 A_{rad} \approx 0.15$ W/cm². For high-density operation, the laser gating is modulated with a duty cycle $D = P_{rad}/P_{laser}$ to preserve thermal equilibrium.

Table 2: Recommended hybrid parameters for Ta_4HfC_5 MULAN v3

Chip area (cm ²)	Laser power (W)	Power density (W/cm ²)	Duty Cycle (4× Radiator)	Average Thrus
1	30	30	0.002	0.0015
10	30	3.0	0.02	1.5
100	30	0.3	0.2	15
400	30	0.075	1.0 (CW)	300

5 Experimental Test Protocol for $K > 1$

To empirically validate the covalent strain and upconversion mechanics:

- Fabricate a Ta_4HfC_5 nanolattice layer topped with upconversion microspheres and a nano-hollander slit layer, enclosing nanoconfined H_2O .
- Fire a pulsed NIR laser source ($\lambda_{in} = 1064$ nm, 10 ns pulse-width, 1 mJ).
- Capture the output radiation profile via an angled spectrometer and a calibrated photodiode array.
- Verify spectral blue-shifting ($\lambda_{out} < \lambda_{in}$), mapping the output wavelength to compute the empirical amplification factor $K = \lambda_{in}/\lambda_{out}$.
- Success threshold is defined at $K > 1.05$ to conclusively clear standard instrument drift and baseline measurement error.

6 Implications for Space Propulsion

- **Zero Propellant Consumption:** The enclosed molecular system undergoes repeatable mechanical strain cycles, acting purely as a momentum-conversion catalyst.
- **Decades-Long Operational Lifetime:** Eradicates fuel depletion, grid erosion, and high-voltage breakdown failure modes.
- **Scalability:** Outperforms electrostatic ion engines at macro-surface scales ($A_{chip} > 0.13$ m²) under identical power constraints.

7 Conclusion

MULAN v3 replaces speculative electronic transitions with a robust thermo-mechanical framework based on geometric nanoconfinement and covalent bond pre-straining. By routing laser triggers through an upconversion and nano-hollander sequence, the system draws heavily from the thermal energy of a refractory Ta_4HfC_5 substrate to achieve an ideal momentum amplification factor $K \approx 15$. This architecture provides a mathematically consistent, non-exotic engineering roadmap toward high-endurance, propellantless space propulsion.

References

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